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Introduction

Natural Language Processing (NLP) is a rapidly advancing subfield of Artificial Intelligence (AI) that focuses on enabling machines to understand, interpret, and generate human language. Among the many challenges in NLP, the detection of figurative language — particularly sarcasm — remains a complex and actively researched problem. Sarcasm is a form of verbal irony in which the intended meaning is typically the opposite of the literal meaning of the words used. It is widely prevalent in digital text, particularly in social media posts, online reviews, and news commentary.

This project presents a machine learning-based system for detecting sarcasm in news headlines. News headlines serve as a particularly interesting domain for sarcasm detection because they are concise, context-dependent, and often written with subtle linguistic cues. The ability to automatically identify sarcastic headlines has significant implications for tasks such as sentiment analysis, news credibility assessment, and public opinion mining.

The proposed system employs classical machine learning algorithms — Naïve Bayes, Logistic Regression, and Random Forest — combined with TF-IDF (Term Frequency-Inverse Document Frequency) feature extraction. The dataset used is the publicly available News Headlines Dataset for Sarcasm Detection (v2) by Rishabh Misra, which comprises approximately 28,619 labeled news headlines sourced from The Onion (sarcastic) and HuffPost (non-sarcastic). [1]

The system is designed to be lightweight, interpretable, and deployable as a standalone prediction module, making it suitable for integration into larger NLP pipelines or web-based applications.

Problem Statement

The proliferation of digital news and social media has created an urgent need for automated tools capable of understanding the tone and intent behind textual content. Sarcasm, by its very nature, is difficult to detect computationally because it relies on contextual knowledge, cultural understanding, and subtle linguistic patterns that are not easily captured by standard keyword-based or rule-based approaches.

The primary problem addressed in this project is the binary classification of news headlines into two categories: sarcastic and non-sarcastic. The challenges inherent to this problem include:

- The absence of explicit linguistic markers that consistently indicate sarcasm.
- Heavy dependence on contextual and world knowledge for accurate interpretation.
- Dataset bias arising from the use of domain-specific sources (The Onion and HuffPost).
- The ambiguity of certain headlines that may appear sarcastic or non-sarcastic depending on the reader's perspective.

Addressing these challenges requires a robust feature extraction strategy and a comparative evaluation of multiple classification algorithms to identify the most suitable model for this task.

Objectives

The objective of this project is to develop a machine learning pipeline capable of automatically detecting sarcasm in news headlines. The system will utilize **TF-IDF vectorization** as the primary feature extraction technique to transform textual data into numerical representations suitable for machine learning algorithms. Three classical classification models—**Naive Bayes**, **Logistic Regression**, and **Random Forest**—will be implemented and evaluated to determine their effectiveness in sarcasm detection. A comparative analysis of these models will be conducted using standard evaluation metrics such as **accuracy**, **precision**, **recall**, and **F1-score** in order to assess their performance. Based on this evaluation, the best-performing model will be selected and saved as a deployable inference module. Additionally, a lightweight and interactive prediction interface will be developed that allows users to input custom news headlines and receive a sarcasm classification along with a corresponding confidence score. The entire project will be structured in a **modular**, **reproducible**, and **platform-independent manner**, ensuring ease of maintenance, version control, and collaborative development.

Literature Review

Sarcasm detection has been an active area of research in NLP for over a decade. Early approaches relied primarily on lexical and syntactic features, such as the presence of interjections, punctuation patterns, and sentiment polarity shifts within sentences. These rule-based methods, while interpretable, demonstrated limited generalizability across different domains and writing styles.

Subsequent research explored the application of classical machine learning algorithms, including Support Vector Machines (SVM), Naive Bayes classifiers, and Logistic Regression, in conjunction with bag-of-words and TF-IDF representations. Studies demonstrated that TF-IDF combined with Logistic Regression yielded competitive performance on short-text classification tasks, making it a suitable baseline for sarcasm detection in brief textual units such as news headlines.[4]

The advent of deep learning introduced more sophisticated approaches. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks demonstrated an improved ability to capture sequential dependencies in text, while Convolutional Neural Networks (CNNs) proved effective at extracting local n-gram features relevant to sarcastic expressions.

More recently, transformer-based pre-trained language models such as BERT (Bidirectional Encoder Representations from Transformers) have achieved state-of-the-art performance on sarcasm detection benchmarks[6]. These models leverage large-scale pre-training on diverse corpora and fine-tuning on task-specific data, enabling them to capture nuanced semantic and contextual information that earlier models could not.

The dataset used in this project, introduced by Rishabh Misra, has become a widely adopted benchmark in the sarcasm detection literature[1], with reported classification accuracies ranging from approximately 80% for classical ML models to over 90% for fine-tuned transformer architectures. This project contributes to this body of work by implementing and comparatively evaluating three classical ML classifiers on the version 2 release of this dataset.

Methodology

The methodology adopted in this project follows a standard supervised machine learning pipeline, comprising the following stages:

1 Data Collection

The dataset used is the News Headlines Dataset for Sarcasm Detection (Version 2), originally compiled by Rishabh Misra [1]. The dataset consists of 28,619 news headlines, each labeled as either sarcastic (1) or non-sarcastic (0). Sarcastic headlines were sourced from The Onion, a satirical news outlet, while non-sarcastic headlines were collected from HuffPost. The dataset is well-balanced, containing 13,634 sarcastic and 14,985 non-sarcastic entries, with no missing values.

2 Data Preprocessing

Prior to model training, the raw headline text undergoes a series of preprocessing steps to reduce noise and standardize the input:

- **Lowercasing:** All characters are converted to lowercase to ensure uniformity.
- **Punctuation and Number Removal:** Non-alphabetic characters are removed using regular expression substitution.
- **Whitespace Normalization:** Leading and trailing whitespace is stripped from each headline.

It is deliberately noted that common stop words such as 'not', 'never', and 'actually' are retained, as these carry semantic significance in the context of sarcasm detection.

3 Feature Extraction

Text features are extracted using TF-IDF (Term Frequency-Inverse Document Frequency) vectorization. The vectorizer is configured with a maximum vocabulary size of 5,000 features and an n-gram range of (1, 2), capturing both unigrams and bigrams. This allows the model to consider not only individual words but also two-word contextual phrases that may be indicative of sarcastic tone.

The vectorizer is fit exclusively on the training data and subsequently applied to transform both training and test sets, thereby preventing data leakage.

4 Train-Test Split

The dataset is partitioned into a training set (80%) and a test set (20%) using stratified random splitting with a fixed random seed (`random_state = 42`) to ensure reproducibility. This yields 22,895 training samples and 5,724 test samples.

5 Model Training

Three classification algorithms are trained on the TF-IDF feature matrix:

- **Multinomial Naive Bayes:** A probabilistic classifier well-suited to text classification tasks involving high-dimensional, sparse feature representations.
- **Logistic Regression:** A linear model that estimates class probabilities using the logistic function, configured with `class_weight='balanced'` and `max_iter=1000` to accommodate convergence on high-dimensional data.
- **Random Forest Classifier:** An ensemble method that constructs multiple decision trees and aggregates their predictions via majority voting, configured with 100 estimators and `class_weight='balanced'`.

6 Model Evaluation

Each trained model is evaluated on the held-out test set using the following metrics: overall accuracy, per-class precision, recall, and F1-score. A comparative analysis is conducted to determine the most suitable model for deployment.

7 Model Persistence

The best-performing model and the fitted TF-IDF vectorizer are serialized to disk using Python's pickle module, enabling inference without retraining. Relative file paths are employed throughout to ensure cross-platform compatibility.

Tools & Technologies

Tool / Technology	Purpose
Python 3.x	Primary programming language for data processing, model training, and inference.
pandas	Data loading, manipulation, and exploratory analysis of the dataset.
re (RegEx)	Text cleaning via pattern-based substitution of non-alphabetic characters.
scikit-learn	TF-IDF vectorization, train-test splitting, model training, and evaluation metrics [2].
pickle	Serialization and deserialization of trained models and vectorizers.
Spyder IDE	Development environment used for iterative code execution and variable inspection.
Git / GitHub	Version control and collaborative code management.
Streamlit (Planned)	Web-based user interface for interactive sarcasm prediction.

Expected Outcomes

The project is expected to deliver a **fully functional sarcasm detection pipeline** capable of classifying news headlines with an accuracy in the range of 80–84% using classical machine learning models. It will also include a **comparative performance report** that analyzes the strengths and limitations of **Naive Bayes, Logistic Regression, and Random Forest classifiers** for the task of sarcasm detection. The system will produce **serialized model artifacts**, such as `nb_model.pkl` and `vectorizer.pkl`, which allow the trained model to perform real-time inference without requiring retraining.

Additionally, the project will feature an **interactive command-line prediction interface** that accepts user-provided news headlines and returns a sarcasm classification along with a corresponding confidence score. The entire implementation will be organized into a **modular and well-documented codebase**, hosted on GitHub, ensuring that the project is easy to replicate, maintain, and extend for future development. Furthermore, the final version of the project will include a **Streamlit-based web application** that provides a graphical user interface, enabling users to conveniently test and interact with the sarcasm detection model through a web-based platform.

It is further anticipated that the project will highlight the practical limitations of models trained on domain-specific datasets and encourage future exploration of more generalized and context-aware sarcasm detection approaches.

Applications

The sarcasm detection system developed in this project has a broad range of practical applications across multiple domains:

1 Sentiment Analysis Enhancement

Standard sentiment analysis tools frequently misclassify sarcastic statements, as the literal sentiment expressed is opposite to the intended meaning. Integrating sarcasm detection as a preprocessing step can significantly improve the accuracy of downstream sentiment analysis systems.

2 News Credibility and Misinformation Detection

Automated identification of satirical or sarcastic news headlines can assist fact-checking platforms and news aggregators in distinguishing genuine news from parody or satire, thereby contributing to efforts to combat misinformation.

3 Social Media Monitoring

Brands, government agencies, and research organizations that monitor public sentiment on social media platforms can leverage sarcasm detection to obtain more accurate representations of public opinion, avoiding misinterpretation of ironic or sarcastic posts.

4 Conversational AI and Chatbots

Dialogue systems and virtual assistants can benefit from sarcasm detection to improve their contextual understanding and generate more appropriate and contextually aware responses.

5 Content Moderation

Online platforms can utilize sarcasm detection as part of automated content moderation pipelines to better understand user intent, particularly in contexts where sarcastic language may be used to circumvent standard keyword-based filters.

Conclusion and Future Work

This project presents a systematic approach to the automated detection of sarcasm in news headlines using classical machine learning techniques. By applying TF-IDF feature extraction in conjunction with three classification algorithms — Multinomial Naive Bayes, Logistic Regression, and Random Forest — the system achieves competitive classification performance on the News Headlines Dataset for Sarcasm Detection (v2).

Experimental results indicate that Naive Bayes and Logistic Regression yield comparable accuracy in the range of approximately 83%, while Random Forest achieves approximately 79-80%. These results are consistent with findings reported in related literature for classical ML baselines on this dataset [5]. The marginal superiority of Naive Bayes and Logistic Regression over Random Forest is attributed to the high-dimensional, sparse nature of TF-IDF feature vectors, which are particularly well-suited to linear and probabilistic classifiers.

A key limitation identified in this study is dataset bias: the training corpus is drawn exclusively from two news sources with distinct stylistic conventions, which constrains the model's generalizability to out-of-distribution inputs. Headlines that do not conform to The Onion or HuffPost writing styles may be misclassified with moderate to high confidence.

Future Work

Several directions can be explored to further enhance the performance and applicability of the proposed sarcasm detection system. One potential improvement involves the adoption of **deep learning models**, such as Bidirectional LSTM or CNN-based architectures, which are capable of capturing sequential relationships and local n-gram dependencies in textual data more effectively than traditional TF-IDF representations. Additionally, the use of **transformer-based language models**, including BERT or RoBERTa, could significantly improve performance through fine-tuning [6] on the sarcasm detection task, potentially yielding accuracy improvements of approximately 7-10% over the current baseline.

Another important direction involves **dataset diversification**, where the training corpus can be expanded with sarcastic and non-sarcastic headlines from a wider range of news sources in order to reduce domain-specific bias and improve model generalization. The incorporation of **contextual features**, such as article body text, publication source metadata, and social media engagement signals, may further enhance the model's ability to capture subtle indicators of sarcasm beyond headline-level information.

From a system deployment perspective, the project can be extended by fully deploying the **Streamlit-based interface** as a publicly accessible web application, enabling real-time sarcasm detection for end users. Finally, **systematic hyperparameter optimization**, such as cross-validated grid search, can be employed to fine-tune model parameters and further improve generalization performance

References

- [1]. R. Misra and P. Arora, "Sarcasm Detection using News Headlines Dataset," AI Open, Elsevier, 2023.
- [2]. F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.
- [3]. A. McCallum and K. Nigam, "A Comparison of Event Models for Naive Bayes Text Classification," AAAI Workshop on Learning for Text Categorization, 1998.
- [4]. T. Joachims, "Text Categorization with Support Vector Machines: Learning with Many Relevant Features," European Conference on Machine Learning (ECML), 1998.
- [5]. A. Joshi, P. Bhattacharyya, and M. J. Carman, "Automatic Sarcasm Detection: A Survey," ACM Computing Surveys, vol. 50, no. 5, pp. 1-22, 2017.
- [6]. J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," NAACL-HLT, 2019.

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